ENTERPRISE RELEASE HANDOFF v1.0.0-PROD

ComplianceHub.ng

Complete Application Quality Assurance & End-to-End Test Plan

TARGET ENVIRONMENTS	DATABASE CLUSTER	STORAGE VAULT	REGULATORY MANDATE
Vercel (Web) + Render (API)	Neon Lakebase PostgreSQL	Cloudinary Secure Vault	CAMA 2020 & 2026 Tax Act

01 Architecture, Topologies & Pre-Configured Test Personas

ComplianceHub.ng employs a decoupled architecture consisting of a Next.js 16 frontend workspace and a high-throughput Go API backend communicating with a multi-tenant Neon Lakebase PostgreSQL instance and Cloudinary Evidence Vault.

Frontend Topology (Next.js 16 Turbopack)

- **Local Dev URL:** `http://localhost:3000` (or :3003)
- **Routing Style:** Next.js App Router with `trailingSlash: true`
- **RBAC Guard:** RoleGuard context with dynamic role emulation
- **Entity Switcher:** Floating Temporary Entity Switcher Bar

Backend Topology (Go 1.23+ Core API)

- **API Port:** `:8080` (dynamic Render `$PORT`)
- **Health Check:** `/api/health` and `/health` (Status 200)
- **Neon Postgres:** Auto-scaling serverless pool (AWS us-east-2)
- **Cloudinary Vault:** S3-compatible evidence storage (`yugcjpau`)

Pre-configured Test Personas (Multi-Role Emulation)

ROLE	PERSONA NAME	ORGANIZATION / ENTITY	PLAN TIER	PRIMARY COCKPIT
BUSINESS	Adaeze Nwosu	PayWave Technologies Ltd (RC-1894231)	Professional	`/business/dashboard`
EXPERT	Tunde Bakare, CAMS	Bakare Compliance Consultancy	Practice Plus	`/practice/dashboard`
AGENCY	Chiamaka Eze	Apex Compliance Advisory Partners Ltd	Agency Growth	`/practice/dashboard`
STAFF	Babajide Sanwo	ComplianceHub Support Operations	Staff Console	`/staff`
ADMIN	Dr. Olumide Adeleke	ComplianceHub Technologies Nigeria Ltd	Super Admin	`/admin/dashboard`

Pre-loaded Sandbox Nigerian Entities

ENTITY NAME	RC NUMBER	INDUSTRY & SECTOR	JURISDICTION	INITIAL READINESS
PayWave Technologies Ltd	`RC-1894231`	FinTech & Digital Payments	Lagos State	84% READINESS
Prime Haulage & Logistics Ltd	`RC-1423891`	Supply Chain & Interstate Haulage	Rivers (Port Harcourt)	68% READINESS
Swift Health HMO Ltd	`RC-1928340`	Healthcare & Managed Care HMO	FCT Abuja	92% READINESS
Zenith Retail Chains Ltd	`RC-1102938`	Consumer Retail & DNFBP Supermarkets	Lagos State	74% READINESS

02 Module Flow 1: Public Portal, Marketing & Regulatory Intel

Tests public routes, responsive navigation, SEO dynamic blog rendering with atomic view count tracking, and statutory assessment calculators.

TEST ID	TEST TARGET	EXECUTION STEPS	EXPECTED VERIFICATION RESULT	STATUS
TC-PUB-01	Landing Page & Hero Navigation	1. Navigate to `/`. 2. Inspect hero section, statistics ticker (₦18.4M Penalties Prevented), and CTA buttons. 3. Click "Start Free Compliance Assessment".	Page loads smoothly; CTA redirects cleanly to `/onboarding`. Top nav sticks on scroll with blur.	☐ Pass
TC-PUB-02	12 Statutory Modules Directory	1. Navigate to `/modules`. 2. Verify all 12 modules appear (CAC CAMA 2020, FIRS Tax Act 2026, NDPC Privacy, AML/CFT, etc.). 3. Check filter pills and card feature tags.	12 modules rendered with correct statutory acts referenced. Active header link "12 Modules" is highlighted.	☐ Pass
TC-PUB-03	Find an Expert Directory	1. Visit `/experts`. 2. Search by filter (e.g., "AML Specialist" or "Tax Attorney"). 3. Inspect verified CAC accreditation badges.	Filtered expert list updates in real-time with verified credentials and retainer hourly/monthly rates in NGN.	☐ Pass
TC-PUB-04	Public Blog & View Tracking	1. Visit `/blog`. Observe view counts on cards. 2. Click on a featured post (e.g. `/blog/cama-2020-cac-annual-returns-deadline/`). 3. Inspect Markdown parsing and Cloudinary hero image. 4. Refresh page or re-check `/blog`.	Single post renders formatted Markdown headings, code blocks, lists; atomic view counter increments in Neon DB without hardcoded values.	☐ Pass
TC-PUB-05	Pricing Matrix & Tiers	1. Visit `/pricing`. 2. Toggle between Monthly and Annual billing cycles. 3. Click "Start 14-Day Free Trial" on Professional tier.	Prices update with annual discount calculation; clicking leads into registration with tier query param preserved.	☐ Pass
TC-PUB-06	Onboarding Assessment Flow	1. Navigate to `/onboarding`. 2. Enter company name, select industry ("FinTech"), state ("Lagos"), employee count. 3. Proceed through 4 statutory questions and submit.	Generates dynamic compliance readiness score (0–100%) and personalized statutory filing calendar preview.	☐ Pass

03 Module Flow 2: Authentication, Tenant Resolution & Entity Switcher

Tests user login, session caching, RBAC role guard redirection, and the new temporary floating entity switcher bar.

TEST ID	TEST TARGET	EXECUTION STEPS	EXPECTED VERIFICATION RESULT	STATUS
TC-AUTH-01	Direct Login & Preset Emulation	1. Navigate to `/login`. 2. Click demo preset "Business Admin (Adaeze)". 3. Submit login form.	Token cached in `localStorage`; user redirected instantly to `/business/dashboard/`.	☐ Pass
TC-AUTH-02	RoleGuard Protected Route Enforcement	1. While logged in as `BUSINESS`, attempt to visit `/admin/dashboard`. 2. Observe redirect response.	User is blocked from admin suite and redirected safely back to `/business/dashboard` with toast warning.	☐ Pass
TC-AUTH-03	Floating Entity Switcher - Drawer Toggle	1. On `/business/dashboard/`, verify top nav header is clean with NO dropdown next to logo. 2. Locate floating bar at bottom center. Click chevron icon.	Upward drawer smoothly animates open showing 4 Nigerian entities with readiness scores and RC numbers.	☐ Pass
TC-AUTH-04	Floating Entity Switcher - Context Mutation	1. Select "Swift Health HMO Ltd (RC-1928340)". 2. Observe dashboard banner and cockpit header.	Tenant context mutates dynamically; cockpit banner updates to Swift Health HMO Ltd, score updates to 92%.	☐ Pass
TC-AUTH-05	Floating Entity Switcher - Minimize & Dismiss	1. Click the "Minimize" icon on the floating bar. 2. Click the "✕" Dismiss button on the minimized bar. 3. Click the corner restore pill to bring it back.	Bar collapses into compact pill; dismissing hides it completely; corner restore pill brings it back immediately.	☐ Pass

04 Module Flow 3: HubBusiness Workspace (Client Operations Cockpit)

Tests all business sub-routes, navigation active state highlighting, deadline tracking, evidence upload, and board report generation.

TEST ID	TEST TARGET	EXECUTION STEPS	EXPECTED VERIFICATION RESULT	STATUS
TC-BIZ-01	Top Header Nav Active Highlight	1. Navigate to `/business/dashboard/`. 2. Observe the "Overview" tab in the capsule pill header. 3. Click through each tab: `HubCalendar`, `HubDocs`, `People & Vendors`, etc.	Current page is clearly highlighted in dark navy (`bg-[#1E293B] text-white font-bold`) with cyan/brand blue icon; trailing slashes handled cleanly.	☐ Pass
TC-BIZ-02	HubCalendar Statutory Deadlines	1. Navigate to `/business/calendar`. 2. Filter by "CAC (CAMA 2020)" and "FIRS (2026 Tax Act)". 3. Click on an urgent deadline (e.g. CAMA Annual Return).	Displays statutory penalties in NGN (e.g. ₦250,000 fine), statutory due date countdown, and required evidence checklist.	☐ Pass
TC-BIZ-03	HubDocs & Cloudinary Evidence Vault	1. Navigate to `/business/documents`. 2. Click "Upload Evidence" button. 3. Select a PDF/PNG file (e.g. Tax Clearance Certificate). 4. Assign authority: FIRS, category: Tax Clearance. 5. Click "Upload & Encrypt".	File uploads to Cloudinary evidence vault with SHA-256 hash stamp; entry appears in document table with "Verified" badge.	☐ Pass
TC-BIZ-04	People & Vendor Sanctions Screening	1. Navigate to `/business/people-vendors`. 2. Inspect Director KYC & AML screening list. 3. Click "Run NFIU / PEP Sanctions Scan" on a vendor.	Returns AML risk rating ("Low", "Medium", "High"), verification timestamp, and compliance certificate generation.	☐ Pass
TC-BIZ-05	AML & Privacy Specialist Cockpit	1. Visit `/business/specialist`. 2. Inspect NDPC Data Protection Audit Filing status. 3. Review CTR (Currency Transaction Report) thresholds under NFIU rules.	Shows NDPA 2023 compliance status, designated Data Protection Officer (DPO) credentials, and audit filing readiness.	☐ Pass
TC-BIZ-06	Board Compliance Report Generation	1. Visit `/business/reports`. 2. Select Quarter: "Q4 2026 Executive Audit". 3. Click "Generate Board Compliance Digest".	Renders comprehensive executive summary breakdown covering CAC, FIRS, NDPC, and PENCOM statutory standing.	☐ Pass

05 Module Flow 4: HubAgents Practice Suite (Expert & Agency Workspace)

Tests multi-client portfolio management, task execution pipelines, commercial retainer billing, and agency team delegation.

TEST ID	TEST TARGET	EXECUTION STEPS	EXPECTED VERIFICATION RESULT	STATUS
TC-PRAC-01	Practice Overview & Client Portfolio Health	1. Switch role to `EXPERT` or `AGENCY`. 2. Visit `/practice/dashboard`. 3. Review active client retainers count and billable hours.	Cockpit displays accredited practice director badge, 14 active clients, monthly recurring retainers (₦3.2M), and pending tasks.	☐ Pass
TC-PRAC-02	Client Detailed Dossier & Filing History	1. Visit `/practice/clients`. 2. Click on "PayWave Technologies Ltd". 3. Inspect CAC Form 2A, annual return filing history.	Opens `/practice/clients/client-paywave/` with complete client corporate records, assigned filings, and audit notes.	☐ Pass
TC-PRAC-03	Work & Tasks Filing Pipeline	1. Visit `/practice/tasks`. 2. Move task "CAC 2026 Annual Return" from "In Review" to "Filed with CAC". 3. Attach statutory electronic receipt.	Task state updates immediately; client notification triggered; milestone reflected on practice dashboard.	☐ Pass
TC-PRAC-04	Commercial & Retainer Fee Invoicing	1. Navigate to `/practice/commercial`. 2. Inspect invoice list and payment status (Paid, Pending, Overdue). 3. Click "Generate Retainer Invoice" for PayWave.	Generates NGN retainer invoice with Nigerian bank account details (Access Bank / Zenith Bank) and VAT line item.	☐ Pass

06 Module Flow 5 & 6: Support Console & Super Admin Cockpit

Tests internal staff ticket resolution, canned statutory snippets, platform metrics, and the Admin Blog CMS.

TEST ID	TEST TARGET	EXECUTION STEPS	EXPECTED VERIFICATION RESULT	STATUS
TC-STAFF-01	Support Console & Statutory Canned Snippets	1. Switch role to `STAFF` and visit `/staff`. 2. Review inbound client tickets. Select ticket on CAMA penalty. 3. Insert standard CAMA 2020 s.377 penalty response snippet.	Support sidebar accurately highlights active route; canned response populates editor with accurate legal citations.	☐ Pass
TC-ADMIN-01	Super Administrator Platform Dashboard	1. Switch role to `ADMIN` and visit `/admin/dashboard`. 2. Verify platform metrics: Total Tenants, ARR in NGN, Database status.	Super admin cockpit displays live Neon Lakebase status (aws-us-east-2), active audit mode badge, and enterprise metrics.	☐ Pass
TC-ADMIN-02	Blog CMS - Authoring & Publishing	1. Visit `/admin/blog`. Click "New Article". 2. Enter title, slug, excerpt, author: "Ifeanyi Felix", category: "CAC & Corporate Governance". 3. Paste Markdown content with headings, bullet points, and citations. 4. Click "Publish Live".	Article persists to Neon DB `blog_posts` table; immediately appears live on `/blog` and `/blog/[slug]` with 0 views initialized.	☐ Pass
TC-ADMIN-03	Companies & Multi-Tenant Directory	1. Visit `/admin/users`. 2. Search by RC number or company name. 3. Toggle tenant plan or inspect audit log.	Displays multi-tenant registry with assigned tenant IDs and subscription tier limits.	☐ Pass

07 Module Flow 7: Backend API, Database & CI/CD Deployment

Verifies Go API server, Neon connection pooling, Cloudinary image security, Vercel ignore-build automation, and Render configurations.

TEST ID	TEST TARGET	EXECUTION STEPS	EXPECTED VERIFICATION RESULT	STATUS
TC-API-01	Go Backend Dual Health Check	1. Execute `curl -i http://localhost:8080/api/health`. 2. Execute `curl -i http://localhost:8080/health`.	Both endpoints return HTTP 200 OK with JSON `{"status":"ok","database":"Neon Lakebase Postgres (connected)"}`.	☐ Pass
TC-API-02	Go Blog API Atomic View Increment	1. Execute `curl -X POST http://localhost:8080/api/blog/{slug}/view`. 2. Query database views column.	Increments view count atomically by 1; handles trailing slashes without 404 or redirect loops.	☐ Pass
TC-DB-01	Neon Lakebase Connection & SSL	1. Verify `DATABASE_URL` uses `sslmode=require`. 2. Run database query through connection pool.	Connects securely using pooled endpoint without connection starvation under concurrent requests.	☐ Pass
TC-MEDIA-01	Cloudinary Vault & Referrer Security	1. Inspect rendered blog cover and author image tags. 2. Verify domain origin is `res.cloudinary.com/yugcjpau`. 3. Verify presence of `referrerpolicy="no-referrer"` attribute.	Images load with zero 403 CDN blocks; no generic Unsplash images anywhere in codebase.	☐ Pass
TC-CI-01	Vercel Ignore-Build Automation	1. Execute `VERCEL_GIT_COMMIT_REF=dev bash scripts/vercel-ignore-build.sh`. 2. Execute `VERCEL_GIT_COMMIT_REF=main bash scripts/vercel-ignore-build.sh`.	Branch `dev` exits with code 0 (Build Canceled); Branch `main` exits with code 1 (Build Proceeded).	☐ Pass
TC-CI-02	Render Backend Build & Run Command	1. Execute `cd backend && go build -o server cmd/api/main.go`. 2. Verify binary execution with `PORT=8080 ./server`.	Compiles with 0 errors; binds cleanly to assigned port; blueprint `render.yaml` verified.	☐ Pass

08 Production Quality Sign-Off & Handoff Verification

This verification sign-off certifies that all flows above have been reviewed against compliance with Nigerian statutory laws (CAMA 2020, Nigeria Tax Act 2026, NDPA 2023) and production stability requirements.

QA Automation Lead

Signature & Date

Lead Fullstack Engineer

Signature & Date

Director of Product & Compliance

Signature & Date